

University of Trento
Master's Degree in Computer Science



**UNIVERSITÀ
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Data Visualization Technical Report

Data Visualization Lab
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1 City Bike Dataset

For this project, our main dataset is the Citi Bike system dataset, a public repository tracking New York City bike trips since 2013.

1.1 Dataset Structure

Released monthly as open-data CSV files, these records log every completed ride, with high-volume months split across multiple files. Each row represents a single trip, offering detailed attributes that characterize the journey.

1.1.1 Original Structure

The original dataset includes the following columns:

- **Ride ID:** A unique identifier for each bike trip.
- **Rideable Type:** The type of bike used for the trip (“classic bike” or “electric bike”).
- **Started At:** A timestamp indicating when the trip started.
- **Ended At:** A timestamp indicating when the trip ended.
- **Start Info:** Information about the station and location where the trip started, structured as:
 - **Station Name:** The name of the starting station.
 - **Station ID:** The identifier of the starting station.
 - **Latitude:** The latitude coordinate of the starting station.
 - **Longitude:** The longitude coordinate of the starting station.
- **End Info:** Information about the destination station, structured equally to Start Info.
- **User Type:** The type of user (“member” or “casual”).

1.1.2 Extracted Features

Additional per-trip features were derived to support the analysis:

- **Trip Duration:** The duration of the trip in seconds, calculated as Ended At – Started At.
- **Trip Distance:** The distance between the start and end stations, computed using the Haversine formula and multiplied by a circuitry factor of 1.3 to better approximate real routes.
- **Day of the Week:** The weekday extracted from the end timestamp, which is used to partition the data since trips are published based on when they ended.
- **Hour of the Day:** The hour extracted from the end timestamp, for the same reason.

Beyond these per-trip features, two aggregated datasets are derived from the cleaned trip data to support station- and route-level analysis:

- **Station Popularity:** for each station and time period, the number of trips starting and ending at that station, obtained by grouping trips by station.
- **Trips Flow:** for each pair of stations and time period, the number of trips travelling between them, obtained by grouping trips by station pair.

1.2 Analysis and Insights

To ensure transparency and reproducibility, the entire analysis is supported by [Jupyter notebook](#) within our repository. The exploratory analysis is based on the February 2025 trip file and focuses on temporal, spatial, and behavioural patterns in bike usage.

Temporal Analysis Temporal analysis shows clear variations in demand depending on the hour of the day and the day of the week. Trips are concentrated during commuting hours, while weekends show different usage patterns associated with leisure activities.

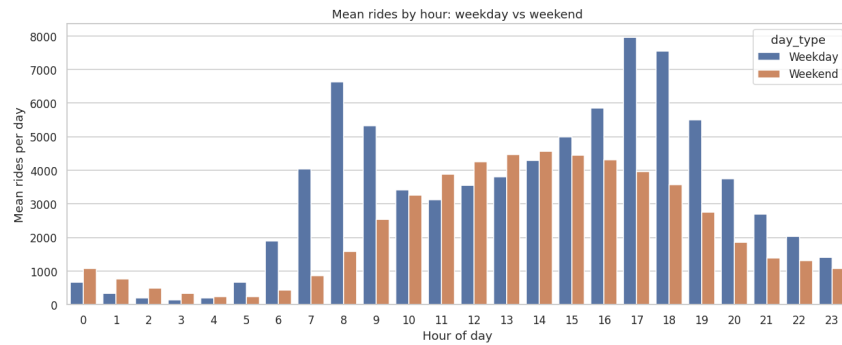


Figure 1: Distribution of trips by hour of the day and weekday/weekend.

Spatial Analysis Spatial analysis highlights the distribution of station origins across the city. Station popularity and trip flows reveal that demand is concentrated around Manhattan and other high-density areas, with some stations acting as important origins.

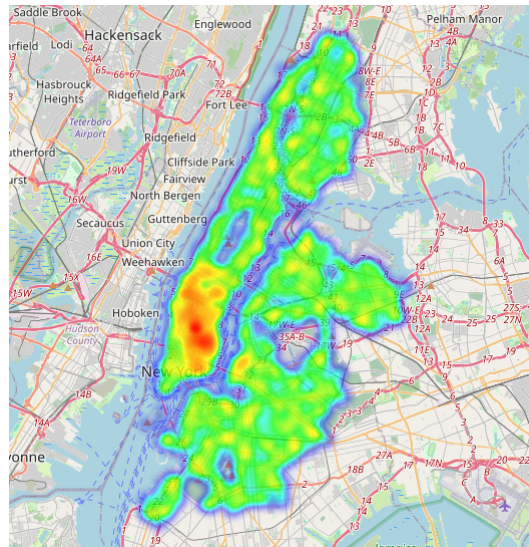


Figure 2: Distribution of trips by station of origin, showed through an heatmap.

User/Bike Type Behaviour Analysis As shown in Figure 3, casual riders lean much more heavily on electric bikes than members do. Furthermore, casual riders take longer trips on both bike types, whereas members maintain a steady, short duration regardless of the bike they choose.

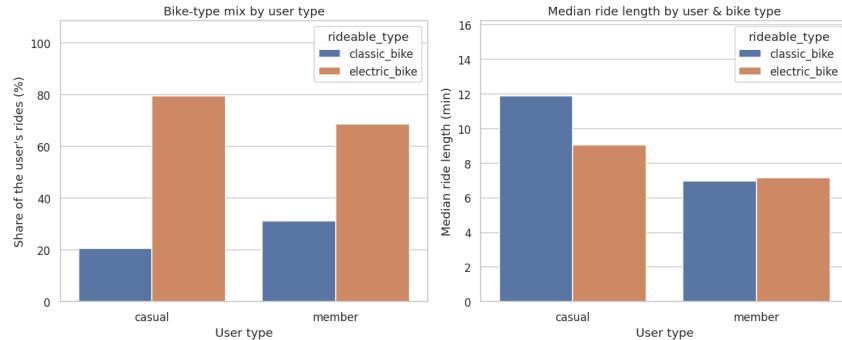


Figure 3: Distribution of trips by user/ride type based on shares of rides and trip length.

Trip Duration and Distance Trip duration and distance distributions provide additional insight into mobility behaviour. Most trips are relatively short, while longer rides are less frequent and often correspond to recreational usage. Durations begin at one minute, as Citi Bike removes trips shorter than 60 seconds at the source. The distributions of trip duration and distance shows some anomalies that we are describing in the following section.

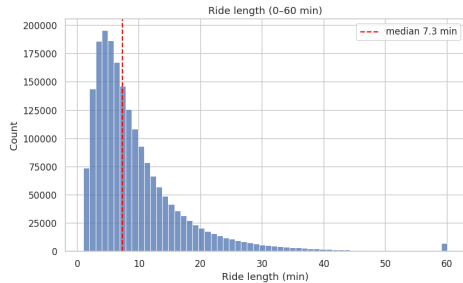


Figure 4: Distribution of trip durations.

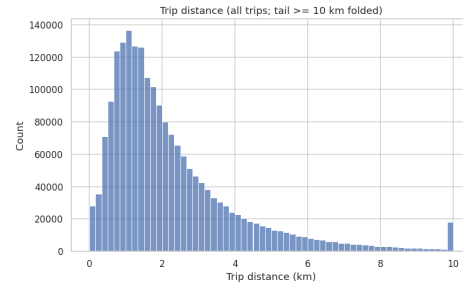


Figure 5: Distribution of trip straight-line distances.

These analyses justify our dashboard proposal, as they clearly reveal patterns in temporal demand, station-level activity, user and bike-type behaviour, and trip duration and distance.

1.3 Data Quality

Various data quality issues were encountered during the analysis of the City Bike dataset.

Different Legacy Structure Around 2020, Citi Bike updated its trip-file schema. Files from the 2013-2019 period use a legacy layout featuring different column names, dropped attributes, and distinct formatting (see the analysis notebook for further details). Despite these discrepancies, this historic data remains highly valuable.

To leverage it, the preprocessing pipeline implements specific schema adjustments, standardizing all monthly files into a single, compatible feature set.

Missing Station Values A small fraction of trips contain missing values, exclusively within the station-related columns. The end-of-trip fields are the most affected: in a representative month, roughly 4,800 trips (0.24%) lack an end station identifier, while start station fields are missing just 0.03% of the time. This issue likely occurs when bikes are abandoned or returned outside a dock. A complete per-column breakdown is available in the analysis notebook. Because valid origin and destination data is essential for spatial and distance metrics, these incomplete rows are dropped during preprocessing.

Trip Outliers Beyond missing values, a small number of trips are clear outliers. In a representative month, about 315 trips (0.02%) last up to 25 hours, appearing in the final bin of the duration histogram (Figure 4). These likely reflect unreturned bikes rather than continuous rides, so trips exceeding a maximum-duration threshold are dropped. Additionally, around 27,000 trips (1.4%) start and end at the same station, resulting in a zero straight-line distance (Figure 5); because distance metrics are meaningless for these round trips, they are removed. Finally, some trips report impossible distances over 11,000 km, as an example, due to corrupted station coordinates, and are filtered out using a maximum-distance threshold.

Biases This dataset carries certain inherent limitations. Because it relies exclusively on the Citi Bike system dataset, the findings may not accurately reflect the demographics of the entire population of cyclists in New York City. Additionally, trip distances are calculated using the Haversine formula between the start and end stations; this straight-line metric serves only as a geometric approximation and cannot account for the actual routes or detours taken by riders. Unfortunately, these structural data constraints cannot be resolved within the scope of our project.

2 Station Metadata Dataset (GBFS)

To enhance the station data provided by Citi Bike, we incorporate the real-time feed published in General Bikeshare Feed Specification (GBFS) format and documented on the Citi Bike System Data page.

2.1 Dataset Structure

Unlike historical trip records, station metadata is served via a public JSON API adhering to MobilityData’s GBFS standard. From this service, we query two key Citi Bike endpoints:

- `station_information.json`: Static properties like location and capacity, whose content changes only when the network itself changes.
- `station_status.json`: Live availability and operational state.

2.1.1 Original Structure

The two endpoints share the `station_id` field, on which they are merged to produce a unified station record comprising:

- **From `station_information`:**

- **Station ID:** internal GBFS identifier, a UUID for most stations and a numeric string for the rest.
- **Short Name:** public station identifier used for joins with Citi Bike trip data.
- **Name:** station name.
- **Latitude** and **Longitude:** station coordinates.
- **Capacity:** total number of docks installed at the station.
- **From station_status:**
 - **Bikes available:** currently rentable bikes.
 - **Vehicle types available:** per-type counts of the bikes currently at the station.
 - **Docks available:** currently available docks.
 - **Operational flags:** `is_installed`, `is_renting`, and `is_returning`.

2.1.2 Extracted Features

Using the three operational flags (`is_installed`, `is_renting`, and `is_returning`), we derive a boolean `active` indicator to filter out unavailable stations from live views.

2.2 Analysis and Insights

The analysis focuses on station capacity, live availability, and docks reconciliation.

Capacity Distribution Figure 6 illustrates network capacity provisioning, which distinguishes smaller stations from major hubs. Nearly half of the stations fall between 20 and 30 docks, meanwhile the bar at 0 docks is not corrupt data: those 42 stations appear in the feed but are not active (demonstrated in the [Jupyter notebook](#)).

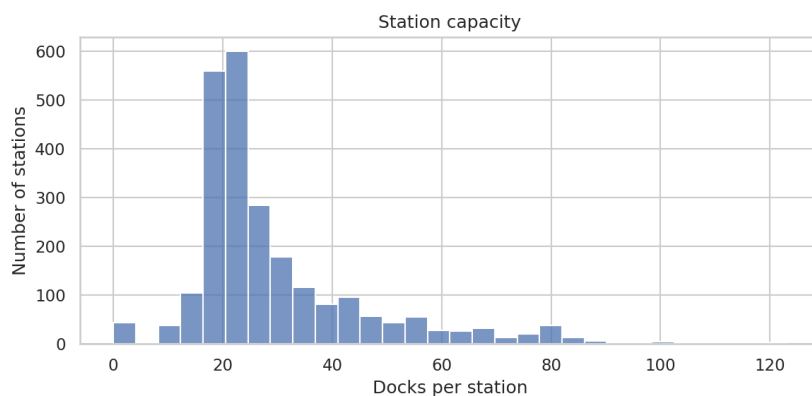


Figure 6: Distribution of station capacity, measured as the number of docks per station.

Live Availability Live bike and dock availability reflects the real-time state of the system. Because the feed is a dynamic snapshot, Figure 7 represents a single

point in time rather than a strictly reproducible state. The chart decomposes total network capacity, classifying every dock into exactly one of six defined states. The same was done for active and inactive stations, for a more granular view.

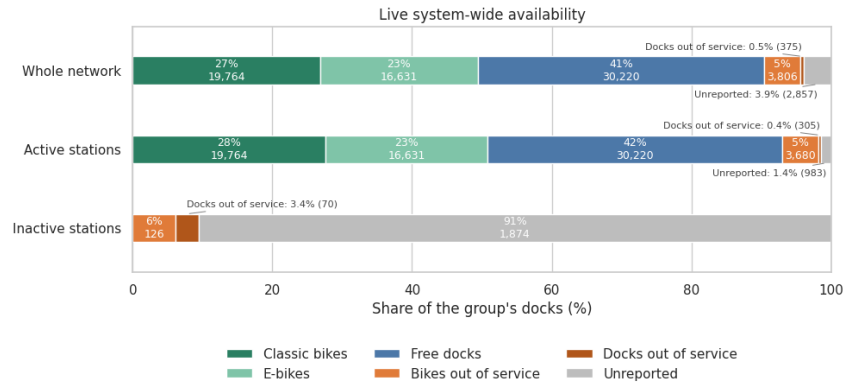


Figure 7: System-wide capacity breakdown across all, active, and inactive stations. Total capacity is fully decomposed into six mutually exclusive states: rentable classic bikes, rentable e-bikes, free docks, out-of-service bikes, out-of-service docks, and unreported capacity.

2.3 Data Quality

The main quality issues concern unreported capacity, the self-consistency of the feed, and identifier consistency. Ultimately, as demonstrated in the notebook, there are no missing values for stations in the feed.

Inactive Stations The feed includes stations being installed, repaired, or removed, all grouped under the `inactive` category. They make up 101 out of 2,462 stations (4.1%) and 2,043 docks (2.8% of capacity). Figure 7 shows why we remove them, since they report capacity but no live counts, and their bikes are out of service. Keeping them would show false empty stations on the map and distort capacity stats without adding real bike availability.

Unreported Capacity The unreported capacity (Figure 7) arises from using two different endpoints. Station capacity is a static property provided by `station_information` and updated periodically, whereas live bike and dock counts are delivered continuously by `station_status`. Figure 8 highlights this discrepancy, showing the per-station gap between declared capacity and the sum of live counters. Filtering out the inactive stations described above eliminates the main source of variance without altering real availability. For any remaining active stations that do not reconcile, dock states are read directly from live counters, treating capacity as a rough benchmark.

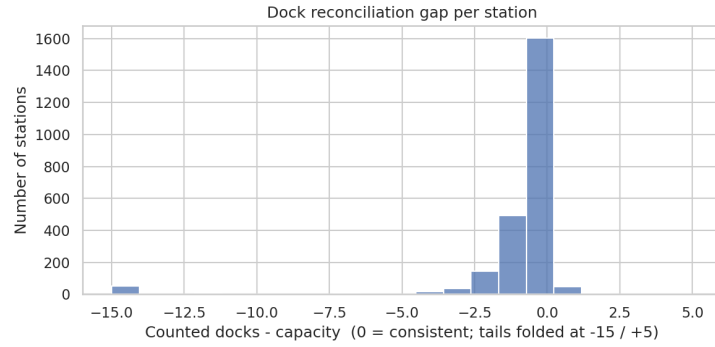


Figure 8: Per-station gap between the declared capacity and the docks accounted for by the live counters.

Identifier Consistency GBFS identifies stations in two ways: an internal `station_id` (either a UUID or a numeric string) and a public `short_name`. Trip history files use only the latter: 93.8% of station references in historical trips match a `short_name`, whereas none match a `station_id`. Therefore, all joins between live metadata and trip history should rely on the `short_name` field.

Live-Only Metadata Citi Bike provides station metadata only as a live feed, with no historical archives. Consequently, removed stations retain their past trip records but lose all feed attributes, while newly added stations lack historical data, making them appear prematurely inactive. This bias only affects feed-specific attributes like capacity and operational status, as each trip record independently contains its own station name and coordinates.

3 Weather Dataset

To enrich the dashboard with meteorological analysis, we incorporate hourly NYC weather data retrieved from the Open-Meteo archive API.

3.1 Dataset Structure

Weather data are queried via the Open-Meteo JSON API by specifying coordinates, a date range, and variables. The API returns parallel hourly arrays derived from reanalysis models that update with a few days' lag, keeping the historical analysis stable.

3.1.1 Original Structure

The dataset includes several attributes, but we consider only the following main fields:

- **Timestamp (time):** hourly observation time, requested in UTC and converted to `America/New_York` on our side so that it aligns with the trip timestamps.
- **Temperature (temperature_2m):** air temperature in °C.
- **Precipitation:** hourly precipitation in millimetres.
- **Weather Code:** WMO weather condition code.

- **Wind Speed (`wind_speed_10m`):** wind speed in km/h.

3.2 Analysis and Insights

Weather conditions serve as a key driver of daily bike usage. Rather than examining weather in isolation, the following section focuses directly on its relationship with overall ridership patterns.

Ridership vs. Weather Figure 9 illustrates daily ridership alongside mean temperature and total precipitation. The data shows a clear inverse relationship between precipitation and usage: the month’s peak precipitation directly coincides with its lowest ridership volume. Meanwhile for the temperature, it’s possible to see also a positive correlation with ridership, where the rides follows the temperature increase. This confirms that weather tracking provides valuable context for interpreting demand fluctuations and supporting operational planning.

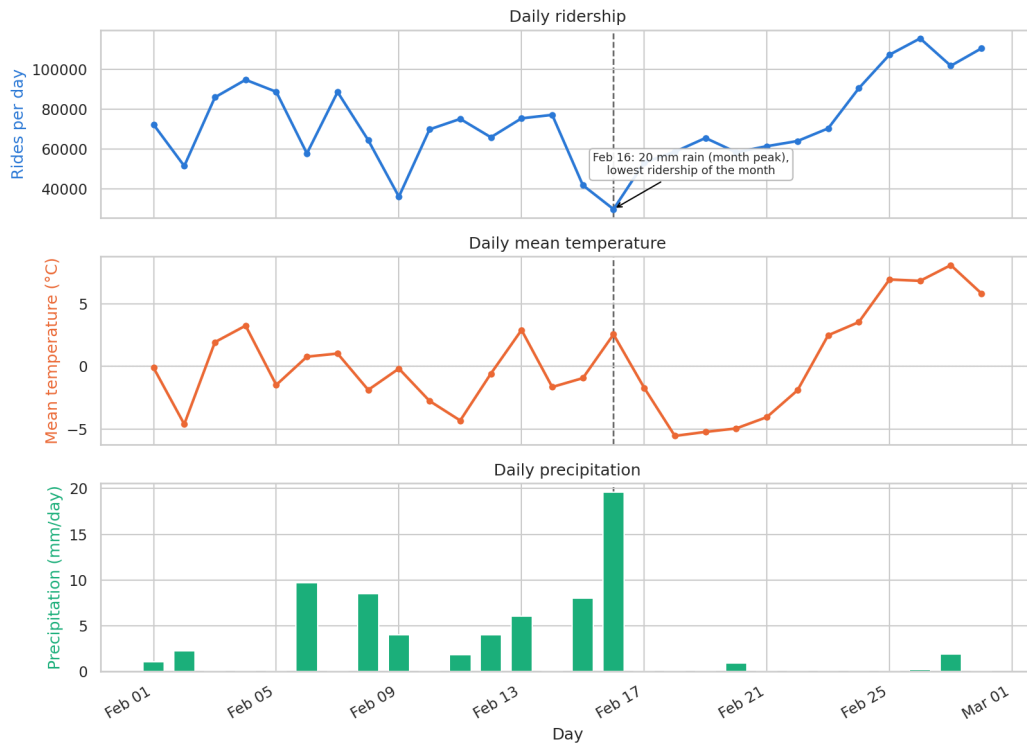


Figure 9: Daily ridership, mean temperature, and total precipitation over the reference month, plotted on aligned date axes with individual scales. The dashed guide highlights 16 February, the month’s wettest day and also its least ridden.

3.3 Data Quality

The archive is complete, with an hourly series covering the entire reference year and no missing values in any variable. The issues we found therefore concern how the values are localised and what they represent, rather than gaps in the data.

Spatial Coverage Weather data is retrieved for a single point centered on Central Park and applied across the entire system. While this dataset misses microclimate variations across a more granular spatial scale, applying higher spatial granularity would introduce more modeling challenges than necessary for a city-scale analysis.

4 Bike Lanes Dataset

To enrich our station and trip insights, we incorporate New York City’s bike-route network, retrieved from the NYC OpenData portal.

4.1 Dataset Structure

Distributed as a single CSV file, the dataset provides a current snapshot of the bike network alongside a historical record, with the NYC Department of Transportation updating the file on an annual basis.

4.1.1 Original Structure

Each record answers four questions about one piece of the network: where it runs, what kind of facility it is, when it existed, and which route it belongs to.

- **Geometry:** the set of lines defined in coordinates tracing the bike route.
- **Street, From Street, To Street:** the street the facility runs along.
- **Borough:** which of the five boroughs the segment lies in.
- **Facility Class:** the type of facility across four classes: *Protected*, *Conventional*, *Shared Lane or Signed Route*, and *Link*.
- **Status:** whether the facility is still in service or has been retired.
- **Installation Date** and **Retirement Date:** when the facility opened and, for facilities no longer in service, when it was removed.
- **Route ID:** the named route the segment forms part of.
- **Segment ID:** which piece of street the record covers. The numbering comes from LION (*Linear Integrated Ordered Network*), the city’s official street map, so every facility is pinned to a known piece of road.

Each facility also records finer direction-specific types alongside its greenway name and managing authority. None of these are loaded into the database, because no view reads them: the thirteen facility types the file distinguishes collapse into only four classes, so a buffered curbside lane and a plain painted one are both *Conventional*, and the greenway attributes describe an axis the visualization does not show.

4.2 Analysis and Insights

The analysis addresses the two questions the dashboard puts to this dataset: what the network looks like today, and how it got there.

The Network Today Figure 10 draws every segment in service, coloured by facility class. The lanes alone redraw the city, which is what makes this layer usable as a spatial backdrop, but the colour carries the actual finding: protection is not distributed evenly. Manhattan is the only borough where *Protected* is the majority class, at 54% of its segments, while Brooklyn sits at 26% and leans on *Conventional* striping instead. Brooklyn, Queens and Staten Island each leave roughly a quarter of their network as *Shared Lane or Signed Route*, which amounts to a marking and a sign rather than a facility. A rider crossing a borough line can therefore drop a protection level without leaving the network, and surfacing that contrast is precisely why the map is coloured this way rather than uniformly.

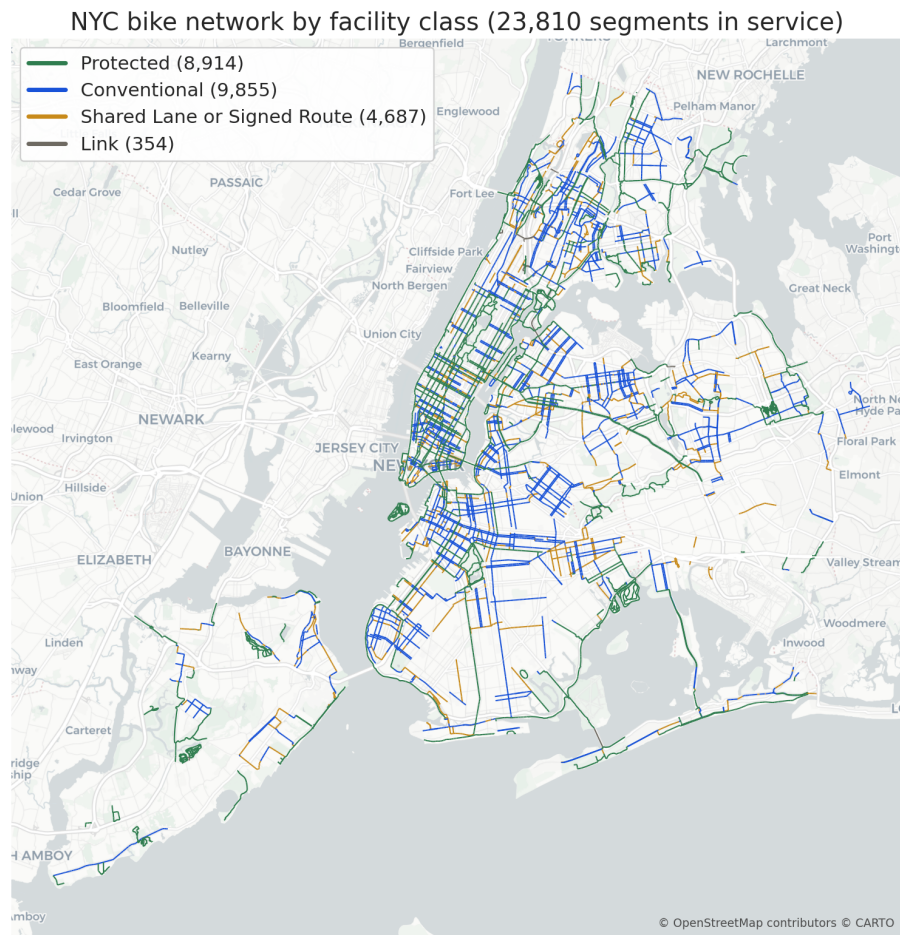


Figure 10: The bike-route network in service, coloured by facility class over the same palette and basemap the dashboard uses. Counts in the legend are segments, not kilometres.

How the Network Grew Because retired facilities are kept, the network can be replayed year by year. Figure 11 counts installations above the axis and retirements below it, the same form the dashboard uses to let the user step back through the network’s history. The build-out is overwhelmingly recent: 21,207 of the 28,983 records date from 2007 onward, whereas the 2,893 predating 1996 are mostly protected off-street infrastructure, greenways, park drives and bridge crossings absorbed into the network rather than built as bike lanes.

The bars below the axis need to be read carefully, and this is the more interesting result. Two thirds of all retirements (66.9%) are matched by a new record on the same stretch of street installed the very same day, and 71% of those come back *better* protected, most commonly a conventional lane rebuilt as a protected one. The red bars therefore measure churn from the upgrade programme far more than they measure lanes being taken away, a reading the data dictionary itself supports when it warns that “retired bicycle facilities do not inherently indicate the removal of a bicycle facility”.

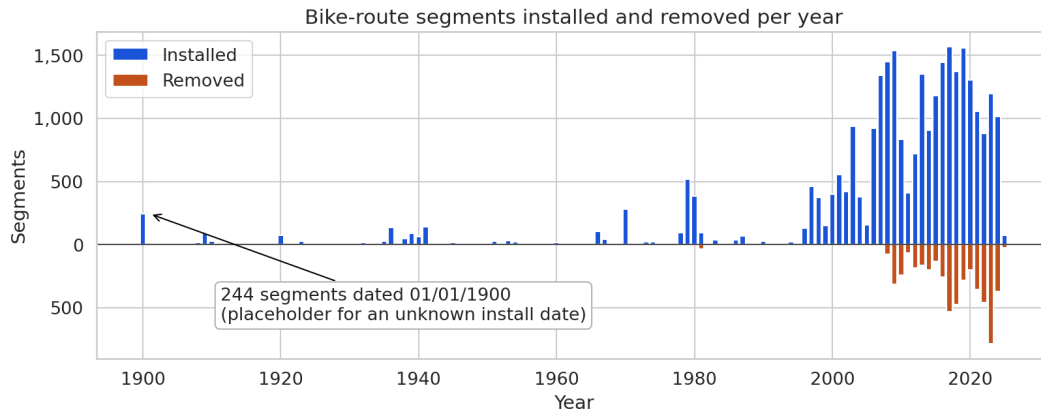


Figure 11: Segments installed and removed per year. Bars below the axis are retirements, two thirds of which are same-day rebuilds of the same stretch of street rather than removals.

4.3 Data Quality

Every column we consume is fully populated, every geometry parses, and none falls outside New York. The issues are therefore not gaps but mismatches between what fields are named and what they contain, all verified in the [Jupyter notebook](#) against the data dictionary.

Identifier Semantics Neither identifier is a row key, and treating either as one is the easiest mistake to make with this file. The Segment ID names a stretch of street rather than a record, so it repeats every time that stretch is rebuilt: 4,603 values carry more than one record, covering 9,489 of the 28,983 rows. The Route ID names a route and is shared along its whole length, with a median of three segments and up to 158, which is what lets the dashboard highlight an entire corridor from a single hovered segment. The attribute that does link a facility to the one it superseded is among those dropped during preprocessing, so that lineage is unavailable downstream. Our database keys rows on the segment, its installation date and its status together, a combination 155 rows still collide on and which are discarded at load.

Sentinel Installation Dates The installation date is never null, which makes the file look complete under a missing-value check, but unknown dates are encoded as the placeholder 01/01/1900 rather than left blank; 244 segments (0.8%) carry it, 196 of them still in service. Because that sentinel is the earliest date in the file, and the year slider derives its lower bound from the earliest installation date, a single placeholder stretches the control across 126 years of which only 67 contain anything.

Status Contradicting the Dates The data dictionary fixes the invariant tying the two together: a current facility carries no retirement date, and a retired one always does. Three records break it, all retired segments on 9th Avenue in Manhattan with no retirement date. As noted in the Dataset Structure section, status never leaves the database, so our year filter is written on the dates alone; these three therefore survive into the present-day network instead of being filtered out with the rest of the retired infrastructure. One further record, on Borinquen Place, is retired two years before its own installation date. The volumes are small enough not to dis-

tort the map, but they are the reason the filter cannot be tightened to treat status and the dates as interchangeable.

Class Is a Maximum, Not a Description Facility class reports “the HIGHEST facility class found along a segment”, so a street with a protected lane on one side and a shared lane on the other is drawn purely as protected. This affects 720 segments (2.5%), which is small enough to leave the map’s colouring unchanged but means it should be read as the best facility available on a segment rather than the facility on both sides of it.